Yiyang Wei

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Education

Zhejiang University — Hangzhou, China
B.Eng in Information Engineering — *Sep. 2020 – July 2024*

- Major GPA(Third Year): 3.72/4
- Selected Courses: Calculus(A+), Linear Algebra(A+), C Programming(A+), Lectures on Programming(A+), Fundamental Data Structures(A), Operating System(A+)

Publications

A Survey of Reasoning-Intensive Retrieval: Progress and Challenges — ACL 2026 (Main)
Yiyang Wei, Tingyu Song, Siyue Zhang, and Yilun Zhao.
Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics, Volume 1: Long Papers, pp. 42101–42122. DOI

Can Retrievers Find the Same Paper from Different Aspects? A Multi-Aspect Full-Paper Scientific Retrieval Benchmark — KDD 2027 (D&B Track)
Yiyang Wei, Fang Guo, Qiji Zhou, Zhizhang Fu, Mengru Ding, Kai Yang, and Yue Zhang.
To appear in the KDD 2027 Proceedings (D&B Track), arXiv version

Selected Work Experience

Westlake University — April 2026 – Now
WestlakeNLP Lab — Research Assistant — *Hangzhou, China*

- **Multi-Aspect Scientific IR Benchmark (KDD 2027 D&B Track):** Led the development of MAPLE, an expert-validated full-paper retrieval benchmark with 2,095 text- and multimodal-referenced queries, and MAPLE-Synth, an OpenReview-grounded retrieval-based ICL pipeline for paper search query generation; evaluated diverse retrievers and identified an 82.4-point gap between AnyAspect@20 (98.1%) and AllAspect@20 (15.7%), highlighting limitations in holistic full-paper representation.
- **Agentic Full-Paper Reranker:** Developing an agentic LLM reranker that directly examines candidate papers using tools (e.g., scientific: paper-structure-aware reader, general: grep/Python), and delegated sub-agents, enabling evidence-grounded comparison across full texts while mitigating information loss from compressed representations.

Yale University — Sep. 2025 – March 2026
Research Collaborator — *Remote*

- **First-Authored Survey(ACL 2026(Main))** A comprehensive survey on Reasoning-Intensive Retrieval (RIR), covering benchmarks, methods across the retrieval pipeline, applications, and open challenges.
- **Synthetic Training Data for Retrievers:** Designed a persona-driven pipeline to construct retriever training data by generating realistic, instruction-rich queries across diverse difficulty levels and styles and mining reasoning-based hard negatives; fine-tuned Qwen3-Embedding on the resulting dataset, improving overall recall by 12% on BRIGHT and achieving state-of-the-art performance on the coding subtask.

Alibaba — July 2024 – July 2025
*DingTalk Wukong(Now **Qwen ToB**) Group — AI Engineer, Full-Time* — *Hangzhou, China*

- **DingTalk AI Image Search:** Fully engineered the foundational online search and offline index construction pipelines. Subsequently optimized the system by deploying a closeness-aware online ranking module and a fine-tuned CLIP offline indexing pipeline, resulting in an 18.9% increase in CTR, a 5% boost in recall, and a 95% reduction in indexing costs.
- **Large-Scale RAG-Based AI Search System:** Led the design of intent recognition and parameter extraction modules, and built **Text-to-SQL**-based user behavior(including send, receive, etc.) retrieval and indexing pipelines, supporting 2M+ users and 100k daily active users, enabling personalized and context-aware search.
- **RAG Evaluation System:** Developed an end-to-end RAG evaluation pipeline that quantified hallucination risk, faithfulness,response-level relevance for (query, retrieval, answer) core triplets in AI Search System.

Zhejiang University — Nov. 2023 – May 2024
Research Intern — *Hangzhou, China*

- Developed a dual-layer federated/pipelined learning framework for LLM fine-tuning on edge devices, reducing accuracy drop to 0.63% (vs. 1.44% in traditional federated learning, a 56% improvement).
- Upgraded communication to gRPC, 2x faster on edge transmission and 40% bandwidth reduction vs HTTP/1.1.

Trip.com April 2023 – July 2023

Intellegence Infrastructure Group — AI Engineer Intern *Shanghai, China*

- Deployed a GPT API service platform handling over 100K daily requests using Spring Boot, SOA, Elastic Search, CAT, and Hystrix for API integration, logging, monitoring, alarming, rate limiting, and load balancing.
- Built an Email-GPT assistant service with Retrieval Augmented Generation (RAG), LangChain, Gradio, and PostgreSQL to parse, translate, and facilitate interaction with international emails, providing writing suggestions

Skills

Programming:	Python, C/C++, Java, Go, SQL
IR Techniques:	Retrieval-Augmented Generation (RAG), Dense Retrieval (BGE-embedding, HNSW, K-means), Sparse Retrieval (BM25, Inverted Index), Multi-modal Search (CLIP, Vision Transformer (ViT))
Framework:	Pytorch, MS-SWIFT, MTEB, LangChain, Flask, Springboot, Gin, Gorm, gRPC
Middlewares:	PostgreSQL, MySQL, Redis, Message queue, Elastic Search, RPC, Sentinel, Hystrix
Developer Tools:	Codex, Claude Code, Git, LateX, Docker, PyCharm, VS Code, IntelliJ, Clion, Goland, Vim, PostMan
Languages:	Chinese Mandarin (Native), English (Fluent, TOEFL iBT 101)